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INFANT LEARNING AND ARTIFICIAL INTELLIGENCE: EXPLORING PROBABILISTIC CONNECTIONS FOR COGNITIVE ADVANCEMENT

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Abstract – There is well-established theory in both infant learning and artificial intelligence (AI) suggesting significant overlap in probabilistic learning mechanisms. Research on infant learning, such as Saffran's (1996) work on transitional probabilities, demonstrates that infants use statistical cues to segment speech and develop word understanding. Similarly, AI models, particularly generative models like neural networks, are trained through statistical methods to simulate human-like cognitive processes (Goodfellow et al., 2016). This paper investigates the intersection of these probabilistic mechanisms, focusing on key studies in infant probabilistic reasoning and the development of language AI models, such as GPT-4.0. The review highlights the shared cognitive functions between infants and AI algorithms through a comparative analysis, emphasizing how both systems predict and process complex inputs from their environments. The discussion moves beyond algorithmic processing to consider the broader implications for advancements in cognitive sciences and AI. By bridging these disciplines, this review positions AI not just as a technology but as a potentially significant model for cognitive development research.

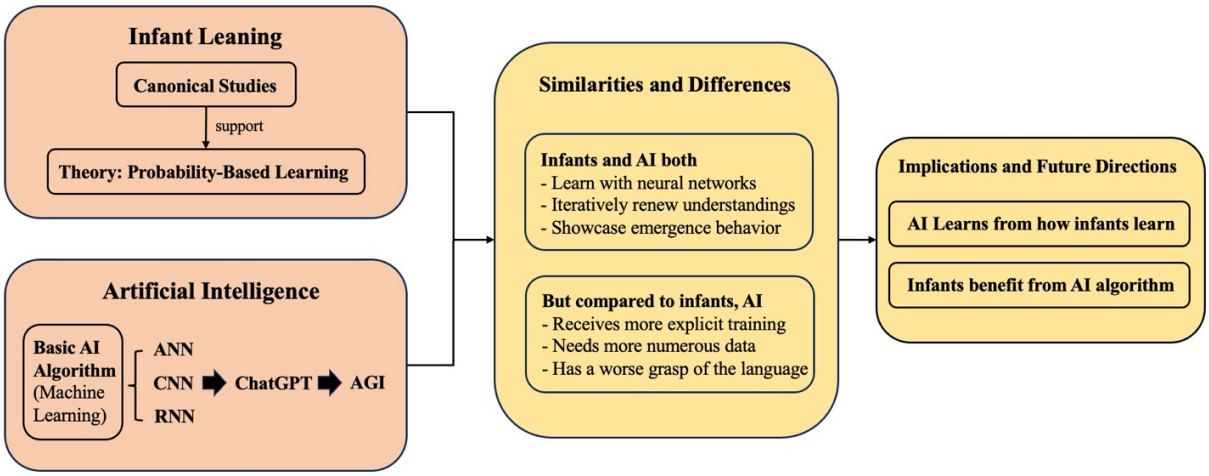
Keywords: infant learning, artificial intelligence, probabilistic learning, cognitive development, generative AI, language acquisition

The method by which infants learn to understand the world differs from that of adults in various ways. Within the domain of language, for example, infants do not immediately grasp the sophisticated vocabulary and intricate grammatical structures mastered by their caregivers. Instead, they develop an understanding of their immediate surroundings by extracting the meanings of individual words from the structures and semantics of the sentences (Wojcik & Saffran, 2015). Here, "structure" denotes the word order in sentences, while "semantics" pertains to the connections between surrounding words. As Saffran (2020) notes, infants discern patterns and regularities; they unconsciously employ a probabilistic method to comprehend language.

Acquiring a similar probabilistic learning technique is Artificial Intelligence (AI), which refers to the capacity of digital computers to perform tasks commonly associated with intelligent beings (Copeland, 2023). Recently, public interest in AI has surged substantially due to Generative AI, which can generate various types of content, including text, imagery, audio, and synthetic data (Goodfellow et al., 2014). The interest in AI continues to grow with the advent of generative AI through Large Language Models (LLM), such as Generative Pre-trained Transformer (GPT) 3.5, which can engage in human-like conversations with remarkable intelligence. GPT 4.0 has garnered increased attention in AI development, and the most recent Sora by OpenAI is a foundational step toward achieving Artificial General Intelligence (AGI) (OpenAI, 2024). Despite its extreme sophistication, generative AI also operates through statistical learning, closely resembling human language acquisition (Dupoux, 2018).

Given that both infants and artificial intelligence systems rely on probabilistic mechanisms to process information, an intriguing question arises: Are AI models imitating the learning processes of human beings? If so, to what extent can these models replicate the sophisticated cognitive functions of early human development? By exploring their similarities and differences, this review aims to assess whether AI models—such as ChatGPT—can replicate or enhance our understanding of human cognitive development (Figure 1). Addressing this question has broader implications, both for advancing AI and for shedding light on how human cognition, especially in early developmental stages, might be better understood and potentially simulated through AI frameworks.

Figure 1.
Conceptual Map: Parallels Between Infant Learning and Artificial Intelligence.



Note. This figure illustrates the logical relationships between various sections of the paper, highlighting the key themes and connections explored in the exploration of infant learning and artificial intelligence.

Infant Learning

Existing Studies

Several significant experiments have demonstrated infants' ability to process probability. Xu and Garcia (2008) meticulously explored how infants could make inferences from samples to populations and vice versa, revealing an innate capacity to process probability. Their experiments involved a controlled setup where infants were shown a population of ping-pong balls of varying colors. They were then shown a series of samples drawn from this population, with some samples accurately representing the diversity of the population and others deliberately skewed. Notably, when the total population was later revealed, infants showed significantly longer gaze durations for the intentionally skewed samples. These results suggest that even at a very young age, infants can predict information about a population (color distribution, in this case) based on their understanding of a sample. Further reinforcing these findings, when infants were briefly exposed to the population before seeing the samples, one representative and the other skewed, infants stared longer at the skewed samples. This indicates that infants used the base rate information obtained from the population to form predictions about which sample was more likely. These reactions were robust indicators of infants' probabilistic processing abilities.

The work of Shultz and Nobandegani (2022) delved deeper into infants' ability to discern probability and predict outcomes. Their experiments echoed the setup of Xu and Garcia (2008), where infants were initially shown the population of objects, followed by samples. The infants' surprise was measurable when presented with skewed samples, suggesting a discrepancy between their predictions and the observed sample. When the sequence was reversed—revealing a disproportionate sample first and the population afterward—the infants' surprise reactions were still prominent, further illustrating their capacity to process both sample-to-population and population-to-sample inferences.

Building on the understanding of infants' skill in handling probability, Saffran et al. (1996) made a significant step toward deciphering infant language acquisition by investigating how infants use probabilities to discern word boundaries. Saffran et al. presented infants with a stream of sounds from an invented language and measured their listening times to both familiar and novel words in that language. The data showed that infants listened longer to novel words, indicating recognition of their unfamiliarity. This suggested that, after merely two minutes of listening, infants could identify the boundaries of unfamiliar words and distinguish them from familiar vocabulary. In the same study, Saffran et al. conducted a complementary experiment where infants were exposed to words and part-words from the same artificial language. Part-words were created by combining the initial syllable of one word

with the final syllable of another. Since part-words included actual word boundaries, they were technically novel to the infants. The infants' longer listening times to these part-words further confirmed that they had learned to identify words as discrete units. This study demonstrates that infants could discern word boundaries solely based on sequential statistics.

Wojcik and Saffran's (2015) study broadened the understanding of infants' statistical learning capabilities by examining their ability to discern regularities within and between sentences. Their findings revealed that infants could recognize not only the "horizontal" relationships of novel nouns that co-occurred within the same sentence but also the "vertical" relationships of nouns based on their consistent positioning within different sentences. The infants could then make more sophisticated predictions based on the regularities they identified across sentences. This ability is foundational for grasping the complex nature of sentence construction and semantics.

Collectively, these experiments illuminate how adeptly infants process probability. From discerning relationships between samples and populations to encoding linguistic regularities, infants demonstrate a remarkable capacity for probabilistic learning. The convergence of these findings underscores the depth of infants' engagement with probabilistic mechanisms, providing compelling evidence of their sophisticated cognitive abilities. As we examine these experiments, it becomes increasingly evident that probability serves as a cornerstone in infants' multifaceted language acquisition journey.

Probability-Based Learning

The experiments conducted empirically consolidate theories regarding infant learning, particularly within the intricate landscape of language acquisition. These studies establish the paradigm of probability-based learning in infants, a concept continuously explored within psychology. This section delves into how infants leverage probability and statistical regularities to acquire language skills.

Acquiring an unfamiliar language poses a formidable challenge for infants, necessitating the initial task of segmenting individual words from the continuous stream of speech (Estes, 2009; Saffran et al., 1996). In essence, infants break down complex sentences into digestible fragments. Once this partitioning is accomplished, the subsequent phases focus on vocabulary (understanding individual words) and grammar (connecting words to make sense of the sentence).

Mastering vocabulary is a process shown to be both statistical and probabilistic, playing a pivotal role in infants' linguistic journey. Toddlers showcase a remarkable ability to infer the meaning of novel words encountered within sentences, demonstrating their adeptness in discerning repeating sounds and predicting subsequent words based on the regularities of fluent language spoken to them (Saffran, 2020; Wojcik & Saffran, 2015). For example, infants can use word order and argument structure to deduce the meaning of a novel verb (Naigles, 1990) and employ familiar verbs to anticipate the semantic properties of upcoming nouns (Wojcik & Saffran, 2015). Through discerning statistical regularity, infants extract unknown words from continuous speech and link them to referents (Estes, 2009; Saffran, 2020). Consequently, infants begin to understand words by their surrounding words, relative position in the sentence, phonotactic patterns, etc. With these probability cues, infants form a primitive understanding of each word. Then, learning becomes iterative: when the word is heard again, infants compare their initial understanding with the new contextual usage (probability cues within the new context). This process enables infants to subtly adjust and refine their comprehension of vocabulary over time (Arciuli & Monaghan, 2009). With each exposure and interaction, infants iteratively fine-tune their understanding until it aligns sufficiently with the context, gradually expanding and sharpening their lexicon (Arciuli & Monaghan, 2009). In essence, infants learn vocabulary through continuous refinement, gradually shaping their rudimentary lexicon to better suit the nuances of their linguistic environment.

The study of infant grammar learning is notably focused on Artificial Grammar Learning (AGL), a paradigm in cognitive psychology and linguistics that evaluates subjects' ability to learn constructed grammar in a laboratory setting (Miller, 1963; Pothos, 2007). In a recent study on AGL, Gervain et al. (2020) generalized infants' ability to learn abstract grammatical and structural properties of language. They argue that young infants can learn simple repetition-based structures, discriminate them from other structures, and discern inconsistencies in unfamiliar structures. Meanwhile, infants identify functors—words whose purpose is to signal grammatical relationships rather than lexical meaning—as anchor points for structural signals that mark parts of speech. As infants detect these grammatical regularities, their expectations of grammatical structure and what they hear allow them to continuously modify and refine their understanding of grammar (Gervain et al., 2020; Saffran, 2020).

In conclusion, infants employ statistical methods to acquire the structural patterns of their native language. They establish and refine associations between sounds and meanings through iterative exposure to linguistic input (Arciuli & Monaghan, 2009). This early stage of infant language acquisition bears striking similarities to the learning processes observed in artificial intelligence when acquiring human language.

Artificial Intelligence

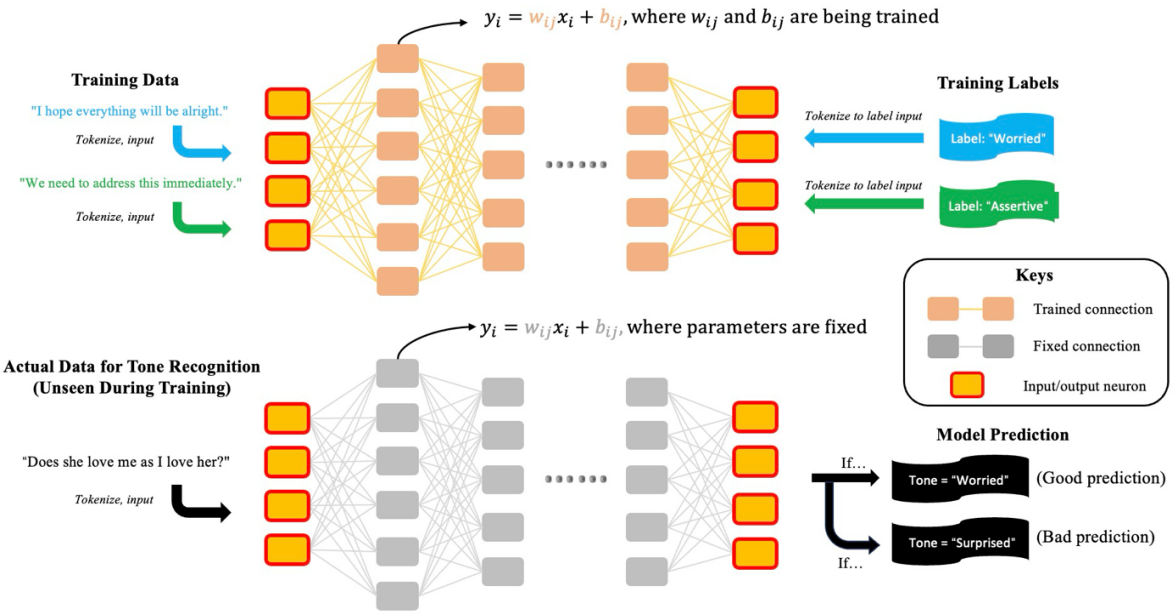
AI Algorithm and Existing AI Models

Intelligence has been defined as the ability to adapt to one's surrounding environment and effectively handle new situations (Abbate, 2023; Anastasi, 1986). Recent advancements in generative AI have progressively impressed the public with the intelligence showcased by AI algorithms. Achieving AI capabilities comparable to GPT has been a gradual process characterized by a learning trajectory. This journey, known as machine learning, involves extracting meaningful representations from data across various levels of abstraction and subsequently optimizing them (Wang et al., 2023). This learning process is fundamentally statistical, akin to how humans acquire language, albeit with a significantly larger volume of data input (Dupoux, 2018).

Machine learning models come in various types, applications, and levels of complexity, but most employ iterative epochs to refine estimations. In language learning, machines often leverage Artificial Neural Networks (ANNs) to enhance performance. Figure 2

Figure 2.
Artificial Neural Network Architecture for Tone Recognition

(a) Top Figure, the model during the training phase.
(b) Bottom figure, the model in practical use.



Note. The model receives tokenized sentences as input, with each box representing a neuron conducting linear regression. Neurons within hidden layers receive input from the preceding layer, perform linear regression, and transmit predictions to the subsequent layer. This process continues until the results reach the output layer, where a final prediction vector is formed. Tokenized labels are provided to the output layer, which compares them with the final prediction. Through training epochs, the model adjusts weights and biases for each neuron to optimize the final prediction's alignment with the given labels.

Once the model is trained, the neurons retain their configurations. When presented with new data, the model endeavors to predict the tone of the sentences. High-quality predictions and elevated accuracy signify a well-suited model, whereas poorer predictions indicate areas for improvement.

depicts the general workings of an ANN, using an example of tone recognition. Sentences in human language are tokenized, establishing a one-to-one correspondence between words and arrays of binary numbers, making abstract sentences understandable for machines (Mikolov et al., 2013). During the learning phase, tokenized labels are assigned to each input sentence. These labels can encompass aspects such as one (joyful, sarcastic, formal, etc.), topic (psychology, cooking, gossip, etc.), or the context (conversation, speech, scientific paper) of the sentence. During training, the ANN seeks to establish (typically linear) relationships in a multidimensional space to connect the input sentence with its corresponding label (Goodfellow et al., 2016; Figure 2a). A network of neurons is deployed to construct a model capable of capturing the intricate relationship between words and their assigned labels. With each training epoch, the weights and biases of neurons undergo adjustments to minimize errors, striving for a better fit (Goodfellow et al., 2016). Once training is complete, the model can be evaluated with unseen sentences to assess its performance in predicting the assigned label (Figure 2b).

This ANN acts as a black box: the model seeks only to establish relationships and does not truly understand the language. Even with high prediction accuracy, it remains difficult for users—and even experts—to explain why a specific set of weights and biases models the data correctly (Zhang et al., 2016). To reduce this uncertainty, more advanced machine learning models for language processing include algorithms such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). CNNs are particularly effective at identifying and processing spatial features, making them ideal for handling visual data, such as images. Each image can be thought of as a grid of pixels, each containing color values for red, green, and blue. CNNs work by applying filters that highlight different aspects of the image. For instance, when distinguishing between images of humans and horses, a CNN might use filters that emphasize the number of legs. A filter that identifies four vertical structures might indicate a horse, while one that detects two might suggest a human. This ability to recognize distinct spatial features is key to how CNNs classify images.

RNNs, in contrast, excel at analyzing data where sequence or order is important, such as time series data or text. They work by processing sequences one element at a time, maintaining a memory of what has been processed so far to use in making decisions about the current element. This is particularly useful in language processing. For example, in understanding a sentence, an RNN processes each word in the context of the preceding words, allowing it to interpret the sentence with an awareness of the evolving meaning from start to finish.

While advanced AI models, exemplified by the transformer model used in GPT, incorporate many complex techniques, the fundamental building blocks often include CNN and RNN architectures and logic (Kim, 2014; Lipton et al., 2015). At the heart of the transformer model is a sophisticated mechanism known as "multi-head attention" (Vaswani et al., 2017). This system enables a dynamic and context-aware analysis of sequences, functioning like a sophisticated interrogation process where each word in a sentence simultaneously queries every other word to understand its function and relationship within the text. Each query gathers distinct "answers" or features, similar to the way filters in CNNs identify relevant spatial features in visual data. These accumulated features are layered, much like filters in a CNN, refining and enhancing the contextual understanding of each word. By processing all words concurrently and retaining their interrelations over time, transformers integrate a temporal dimension reminiscent of RNNs' handling of sequential data. Consequently, despite their simplicity, CNNs and RNNs are foundational components of many sophisticated AI models today.

While advanced model architecture is crucial, the breadth and depth of the training data are also essential in shaping AI's intelligence. The Large Language Model (LLM) plays a significant role in developing adaptable, general language systems (Brown et al., 2020). LLMs analyze extensive language pattern data to identify statistical regularities (features) and word order (time dependence) to enhance models' understanding of language vocabulary and grammar. Although most models do not fully understand the language, they recognize correlations. The LLM-trained model then attempts to comprehend input and generate output in the learned language (Brown et al., 2020). GPT is a transformer model that is pre-trained on vast amounts of text data. It has demonstrated exceptional performance across various natural language processing tasks, including question answering, text summarization, language translation, and text generation (Abbate, 2023).

Despite its complex nature, GPT uses a probabilistic concept mechanism that aligns with basic machine learning paradigms, although it is considerably larger and more complex, with numerous add-ons that form the transformer structure (Abbate, 2023). With the expansive data provided by LLMs, the complex model learns to respond in a human-like manner when users provide input to ChatGPT, showcasing not only

intelligence but also adaptability. ChatGPT can even answer open-ended questions for which there is no predetermined answer, such as, "Write me a paragraph about AI as a real friend to a user." In this example, GPT spontaneously generates a creative response, demonstrating its ability to adapt to novel scenarios (OpenAI, 2023).

It is essential to note that, despite its remarkable capabilities, GPT operates based on vast collections of regressions. It does not possess a fundamental understanding of users' input as language itself. Instead, GPT relies on statistical regularities derived from LLMs and algorithms to generate the most probable responses (Brown et al., 2020). This probabilistic nature grants GPT versatility in responding to a wide array of problems, although it may occasionally produce incorrect answers while confidently asserting their validity (Abbate, 2023).

Artificial General Intelligence (AGI)

In the realm of technology, few recent developments have sparked as much debate as Artificial General Intelligence (AGI) (Faraboschi et al., 2023). Although a universally accepted definition of AGI is lacking, its central attribute is intelligence capable of solving almost any task, often with superior performance to humans (Shevlin et al., 2019). AGI aims not only to mimic human thought processes but also to autonomously complete tasks that humans typically face.

The increasing parallels between AI and human learning processes have become a source of both fascination and apprehension in the public sphere (Faraboschi et al., 2023). Concerns arise as AI systems become capable of performing tasks on par with, or even better than, humans. For instance, AGI can exhibit self-reliance by making decisions based on its learned patterns and data with minimal human intervention. In the realm of language acquisition, AGI could conceivably attain a genuine understanding of human language, equaling or even surpassing the comprehension abilities of infants. Current research in AI language acquisition predominantly revolves around "AI learning from humans," an approach in which human logic is encoded into AI algorithms to enhance language processing performance (Johnson et al., 2022). The trajectory of AI development hints at a potential future where AGI might contribute to enhancing language acquisition processes for infants.

Despite the natural proficiency that infants display in learning language, the rapid advancement of AI capabilities suggests that AGI may eventually offer novel methods for language learning. Reflecting on the evolution of AI, the transition from classical programming to machine learning significantly boosted the language learning capabilities of machines. Given the pace of development in current AI models and the breakthroughs they have achieved, another paradigm shift in future models' language learning capabilities seems plausible. This shift could introduce new patterns and methodologies that not only mirror but also potentially surpass human language acquisition capabilities. While no existing model currently approaches this level of intelligence, the prospect is an exciting possibility that may unfold in the not-too-distant future.

Efforts toward AGI are promising, with the release of Sora by OpenAI in February 2024 marking a significant milestone. Representing a fundamental step toward AGI, Sora demonstrates the capability to generate photorealistic videos and comprehend human instructions with unprecedented accuracy (OpenAI, 2024). This level of intelligence and autonomy is both a significant advancement and a source of alarm, primarily because of Sora's potential to disrupt fields such as movie production and advertising. Public apprehension regarding Sora's intelligence surpassing human capabilities is palpable, with concerns about job displacement and the restructuring of modern industries looming large in discussions about the implications of advanced AI technologies.

A canonical threshold of machine intelligence is the Turing test, which determines whether a machine can demonstrate intelligence equivalent to or indistinguishable from that of a human ("Turing Test," 2024). While no AI has yet passed the Turing test (Powell, 2019), the trend suggests that AI is inevitably becoming more human-like. These concerns are valid, and ethical considerations are paramount in the development of AI toward AGI, drawing parallels with Asimov's (2004) laws of robotics.

Overlapping Principles

Similarities Between Infant Learning and AI

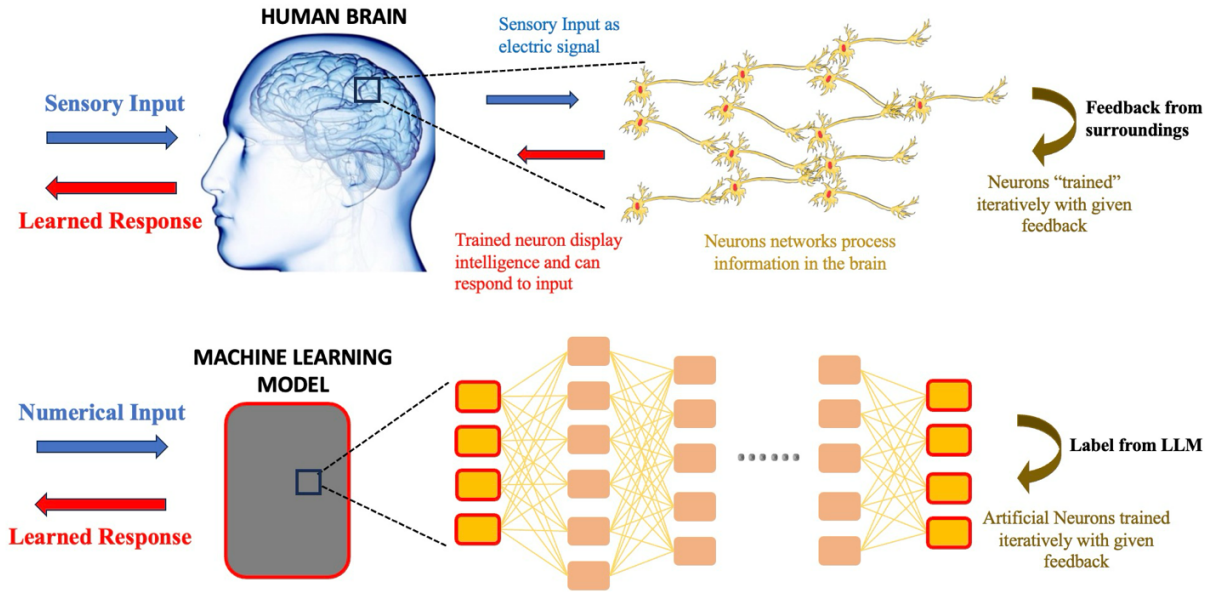
As we explore the processes of infant learning and artificial intelligence, it becomes apparent that their statistical learning mechanisms are closely intertwined (Abbate, 2023; Dupoux, 2018; Wang et al., 2023). Dupoux (2018) characterizes infants and AI as "learners" when confronted with new information and suggests a learning strategy that applies to both human infants and AI: start with simplicity and rely on actual raw data. Since the learning processes of infants and AI align so closely, we attempt to answer the question raised in the introduction: Does machine learning genuinely model infant learning? Before addressing this question, it is essential to note that this paper focuses on the early stages of infant learning, during which infants have not

yet comprehended language. Once infants develop enough understanding to generally comprehend sentences spoken by caregivers and begin explicitly learning the language (akin to an elementary-level English course), they have progressed beyond the scope of this discussion.

When acquiring spoken language, the auditory system transmits sound signals to the brain cortex for processing, where individual neurons function within a sophisticated system that facilitates comprehension (Hickok & Poeppel, 2015). While the intricate workings of neurons and the brain are beyond the scope of this paper, it is crucial to acknowledge the neurons' pivotal role in information processing, enabling humans to derive meaning from linguistic signals (Figure 3a). At this early stage, infants encounter words devoid of understanding, similar to the initial stages of AI grappling with human language. In language acquisition, infants rely on statistical learning mechanisms (Saffran, 2003) to discern structural and semantic regularities within sentences, gradually developing linguistic comprehension by iteratively refining their lexicon (Arciuli & Monaghan, 2009). Similarly, AI demonstrates adeptness at deciphering the semantics embedded within sequential data, including natural language and biological sequences (Wang et al., 2023). Using statistical methodologies, AI acquires proficiency in human language by discerning patterns and regularities through extensive data-driven processes facilitated by neurons in ANN and labels provided by LLM, as shown in Figure 3b (Dupoux, 2018; Goodfellow et al., 2016; Wang et al., 2023). This approach allows AI to grasp aspects of lexicon and

Figure 3.
Comparative Analysis of Human and Artificial Neural Networks

(a) Top figure, human neural network.
(b) Bottom figure, Artificial Neural Network (ANN). Utilizing numerical signals as input, individual neurons within the model collaborate to conduct regressions and generate predictions. Through iterative training by comparing these predictions with labels, the neurons change coefficients in their regression to enhance their responses over time, exhibiting intelligence.



Note. When exposed to sensory input signals, the individual neurons within the human neural network respond accordingly. Through iterative adjustments with contextual cues, the neurons improve their responses over time, represented by an improved understanding of vocabulary, showcasing intelligence. In parallel, individual neurons within the Artificial Neural Network collaborate to conduct regressions and generate predictions for given numerical signals. Through iterative training by comparing these predictions with labels, the neurons change coefficients in their regression to enhance their responses over time, exhibiting intelligence.

grammatical structures inherent in human language, resembling the learning trajectory observed in infants.

The predictive capabilities observed in AI models, in a sense, also mirror the learning process of infants. Both involve processing input information through a neural network, where groups of individual neurons collaboratively perform language acquisition tasks (Goodfellow et al., 2016; Hickok & Poeppel, 2015), albeit with significant differences between the functions of infant neurons and artificial neurons. In infant learning, as outlined previously, infants reassess and refine their understanding of words through iterative encounters. Upon hearing a word previously encountered, they retrieve their initial interpretation and juxtapose it with the word's current usage, considering various probabilistic cues. These cues encompass both vertical and horizontal relationships within language structures, transitional probabilities, and phonotactic patterns, among others. A fine-tuning of comprehension then occurs based on the new context, as described by Arciuli and Monaghan (2009). Grammar acquisition in infants operates under a similar mechanism. Infants compare their pre-existing grammatical framework to new linguistic contexts, adjusting as necessary (Gervain et al., 2020). The process described above resembles one epoch in the machine learning paradigm. As shown in Figure 3a, the neural network of the infant brain, which receives linguistic signals as inputs, processes these signals and generates an output reflecting preliminary comprehension. This output, when confronted with new contextual data (akin to a label in ANN), is subject to revaluation and adjustment, mirroring the process by which an ANN adapts the weights and biases of its neurons to better fit provided labels (see Figure 3b). Occasionally, infants find that their initial understanding is incongruent with the new context—similar to an ANN's initial set of poorly performing coefficients that result in significant loss. However, just as an ANN iterates and optimizes its coefficients in successive epochs, infants also refine their linguistic understandings upon realizing discrepancies in new encounters. Indeed, a parallel can be drawn between the functioning of human neural networks and artificial neural networks, with both involving inputs, predictions, and iterative adjustments based on feedback.

Another notable similarity between infants and AI lies in the phenomenon of emergence. As Wei et al. (2022) argue, emergence occurs when quantitative changes in a system lead to qualitative changes in behavior. In essence, both infants and AI demonstrate intelligence that transcends the information provided in their training data, representing a qualitative shift indicative of genuine intelligence. As infants progress in language acquisition and master vocabulary and grammar, their understanding surpasses probabilities and correlations, marking a significant qualitative change. Similarly, scientists have observed the phenomenon of emergence in various AI models (Wei et al., 2022). For instance, when queried about the possibility of inserting a stick into human ears, GPT surprised researchers by responding affirmatively. It justified its answer by referencing the "golden cudgel" from the Chinese literary work Journey to the West, suggesting that if the stick possessed magical properties described in the literature, it could be inserted into ears. This unexpected connection made by GPT exemplifies emergence, where its extensive training data results in a qualitative change, enabling GPT to draw novel connections and demonstrate intelligence beyond its initial programming (OpenAI, 2023).

Differences Between Infant Learning and AI

Despite the remarkable similarities between the learning processes of human infants and AI, a fundamental distinction arises concerning cognition and its application, primarily in the utilization of statistical techniques (Dupoux, 2018). In other words, the underlying mechanisms of learning for infants and classical AI diverge. However, it is noteworthy that advancements in AI are increasingly aligning with the problem-solving approaches characteristic of human intelligence, similar to those employed by infants.

The process of classical machine learning represented by artificial neural networks (ANN) is strongly supervised, which, as Dupoux (2018) argues, marks a considerable difference from infant learning. Infant learning can be likened to unsupervised learning, where infants are not explicitly provided with a one-to-one correspondence between inputs and outputs but are instead tasked with processing vast amounts of information to derive understanding. This is somewhat analogous to unsupervised machine learning, such as clustering, in which a model is presented with large datasets and tasked with organizing them into clusters in an unspecified manner (Goodfellow et al., 2016). Operating without explicit instructions or labels guiding the model toward a specific task, unsupervised machine learning reflects a process closer to infant language acquisition.

As a result, in language acquisition, infants do not have the advantage that machines do in receiving and comprehending clear and explicit labels (Goodfellow et al., 2016). Machine learning models already excel in the realm of binary language; their task lies in mapping human language to machine language under close

supervision (Lake et al., 2016). In contrast, infants are not born with a pre-existing language, nor do they have another language to refer to when acquiring their first language. Essentially, infants are "on their own" in understanding a new language; they must develop their understanding independently, without pre-existing labels to reference—similar to unsupervised machine learning processes. This process underscores the unique intelligence of human infants.

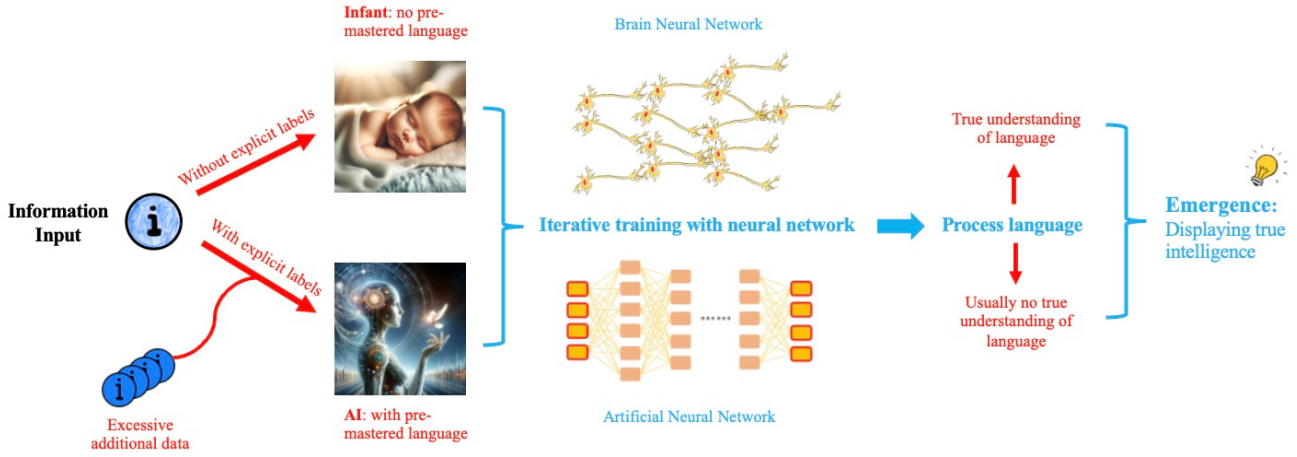
Another notable distinction between human and machine learning processes lies in the volume of data required for learning. Dupoux (2018) highlights that machines typically require around 10,000 hours of learning to accomplish a task that a four-year-old child achieves within 700-4,000 hours. Frank (2023) also observes that the amount of language input for training GPT-3 was on the order of 101210^{12}1012 words, whereas the upper bound of word input for human learners is below 10910^9109. While human learners begin with mere probabilities, their understanding evolves beyond correlations over time, gradually developing a deeper comprehension of language (Denison et al., 2013). In contrast, even after extensive learning, both primitive and advanced AI models often rely on coefficients to determine regressions, lacking the nuanced understanding observed in human learners. Figure 4 summarizes the overall similarities and differences between infant learning and AI training discussed in this section.

It is worth noting that AI faces the daunting task of comprehending a language crafted by human intellect. Dupoux (2018) eloquently outlines three key aspects of human language: its inherent complexity, flexible representations, and capacity as a finite computational system capable of generating an infinite array of expressions—a framework finely attuned to the human brain. These attributes are ubiquitous across all human languages, posing a formidable challenge for AI, which operates within the confines of binary logic. Thus, it is unsurprising that AI might take longer to wrap its circuits around this complex linguistic labyrinth. Attempting to navigate the intricacies of language through brute-force regression in hyperspace would likely prove as futile as trying to teach an infant calculus before they have mastered counting blocks.

Conclusions

This review has examined the nuanced interplay between infant learning and artificial intelligence, focusing particularly on the role of probabilistic learning methods and their implications for language acquisition.

Figure 4.
Comparative Analysis of Language Acquisition in Infants and AI



Infant and AI images were generated using the GPT (Generative Pre-trained Transformer) model developed by OpenAI. OpenAI holds the copyright for the model. Permission to use GPT was obtained through OpenAI's usage policies and terms of service.

Note. This figure contrasts the learning pathways of infants and artificial intelligence in language development. Red arrows and text highlight the differences, emphasizing the absence of pre-mastered language and the reliance on unsupervised learning in infants, contrasted with the binary language comprehension and supervised learning in AI. Blue arrows and text indicate similarities, including the iterative nature of learning via neural networks and the emergence of complex language understanding. This graphical representation underscores the intricacies and nuances in the learning processes of two fundamentally different yet strikingly similar learners.

Our analysis reveals that both infants and artificial intelligence systems fundamentally rely on probabilistic mechanisms to process complex stimuli and learn from their environments. This shared approach underscores a fundamental similarity in how learning potentially occurs across biological and artificial systems, suggesting that AI can indeed mimic certain aspects of human cognitive development.

As the boundaries between human cognition and artificial intelligence continue to blur, we stand on the brink of a new era where the synergistic relationship between these fields could catalyze significant breakthroughs. The convergence of infant learning strategies and AI technologies provides not only a unique perspective on the future of cognitive development but also positions AI as a powerful tool that could reshape our understanding of learning and intelligence. This emerging dynamic has profound implications for both cognitive science and technology, setting the stage for future innovations that could redefine the very fabric of human-AI interaction.

It is important to recognize that the theories and methodologies discussed in this paper represent a selection of widely accepted approaches within the fields of infant learning and artificial intelligence. Both fields, however, are rich with diverse and sometimes competing theories that may offer different perspectives on the mechanisms of learning and cognition. While this review has focused on probabilistic learning due to its significant prevalence and support in recent research, the exclusion of alternative theories is not an assertion of their irrelevance but rather a necessary delimitation given the specific scope of this paper. Examining opposing views and alternative theories in detail remains beyond the scope of this review but represents a valuable avenue for future exploration.

Implications and Future Directions

To forecast the future of AI development, it is essential to examine the current landscape of advanced models and discern the direction of their progression. Successful models like GPT have notably departed from solely relying on classical supervised machine learning, as previously discussed. GPT incorporates unsupervised learning components, enabling it to operate autonomously, akin to human infants, without explicit guidance (Brown et al., 2020). Moreover, GPT's remarkable advancement lies in its adeptness at understanding context. Enhanced with various structural components, such as CNN and RNN, and more advanced tools like reservoirs and transformers (Goodfellow et al., 2016), GPT exhibits a capacity to comprehend context effectively, similar to human logic. These features, analogous to human reasoning, enable GPT to demonstrate intelligence when faced with novel challenges (Vaswani et al., 2017). As a result, advanced AI models aspire (perhaps unintentionally) to emulate human learners by developing capabilities such as understanding context and autonomously finding insights without explicit labels (Vaswani et al., 2017). Thus, we see a trend moving from supervised to unsupervised learning, from brute-force regression toward methods resembling human logic, and from purely data-driven approaches toward incorporating elements akin to human intuition.

AI development is currently focused on two key directions. The first direction involves augmenting model complexity by integrating more sophisticated components, allowing the model to address problems with "logical" reasoning similar to that of humans. These extensions enable AI models to gain a better understanding of context and enhance problem-solving capabilities. The second direction is advancing toward unsupervised learning, analogous to infant learning. With greater autonomy, the phenomenon of emergence—a hallmark of intelligence—is more likely to occur. For instance, the latest OpenAI model, Sora, can generate a video of two pirate ships fighting on a cup of coffee (OpenAI, 2024), a scenario that could not have been directly learned from any existing data. This video clip demonstrates both intelligence and emergence during Sora's training. However, the precise mechanisms and conditions for such qualitative changes in learning, and the amount of data required, remain open questions (Wei et al., 2022). Nevertheless, having more high-quality data for training can facilitate this transformation, with the hope that an increase in quantity will eventually lead to a qualitative shift.

As AI advances in language processing, the possibility emerges for AI to provide feedback to human learners, particularly infants. By emulating the reasoning processes of infants, or humans more broadly, certain AI advancements—such as emergence—may offer insights into infant language acquisition. Consequently, developments in AI could deepen our understanding of human behavior and cognition. Future research will likely extend beyond improving AI models that learn from humans to utilizing AI as a tool to study human behavior (Tan et al., 2024). This trend suggests that AI could play a dual role: as a means to enhance our knowledge of human behavior and as a potential aid in optimizing language acquisition methods for infants. This reciprocal feedback loop implies that infants could learn more efficiently from AI, while AI, in turn, gains insights from observing and understanding the language learning

processes of infants. The symbiotic relationship between AI and infant learning holds promise for advancing our understanding of language acquisition and potentially enhancing strategies for infant language learning.

While AI development offers exciting possibilities, it also raises significant ethical concerns. The advent of AI could profoundly reshape societal structures, such as displacing workers by automating tasks more cheaply and efficiently than humans can perform. More profoundly, if AI models were to significantly enhance human learning capabilities, they could lead to feelings of alienation, as the tools we created begin to surpass us in intelligence and instruct us on optimal behaviors and strategies. Such a shift could challenge our understanding of our role and status on this planet. Historically, similar ethical dilemmas have arisen; for example, the cloning of Dolly the sheep sparked a global debate on the limits of scientific exploration, leading to stringent regulations on cloning technologies. The ethical concerns posed by AI are even more consequential, with potential implications for human significance and existence itself. These considerations underscore the necessity for prudence and foresight in AI development, ensuring that technological advancements do not outpace our ability to manage their impacts responsibly.

Despite the ethical concerns associated with AI's advancement, an optimistic perspective might be more productive, viewing AI as a novel lens that reveals a dynamic panorama of possibilities (Epstein et al., 2023). As we enter a new AI era, our vision expands beyond mere technological advancement. In the unfolding narrative of technological evolution, we witness the inception of AI autonomy. Much as humans rely on intuition and insight to navigate life's complexities, AI systems independently parse, assimilate, and analyze data, making decisions based on patterns they discover. This emerging autonomy is likely to grow as AI sophistication increases. Although the nature of human and AI intelligence differs fundamentally, both share an inherent autonomy in engaging with the world. Thus, it is with profound insight that we recognize our partnership with AI in the grand journey of learning and civilization, signifying that humanity, in its quest for knowledge, no longer walks alone.

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